DSA Experiment 2

Title: Perform recursive Function binary search using C.

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```
#include <stdio.h>
#include<stdlib.h>
int BS(int arr[],int l,int r,int target);
int main() {
int size;
printf("Enter size of array:");
scanf("%d",&size);
int arr[size];
int i;
printf("Enter an array:");
for(i=0;i<size;i++)
{
scanf("%d",&arr[i]);
}
int n = sizeof(arr)/sizeof(arr[0]);
int target;
printf("Enter the no. to be searched:");
scanf("%d",&target);
int result=BS(arr,0,n-1,target);
if(result==1)
{
   printf("No. is found!!");
}
else
{
printf("No. is not found :(");
}
return 0;
}
int BS(int arr[],int l,int r,int target)
{
if(l>r)
{
   return 0;
}
```

```
    int mid=l+(r-l)/2;
    if(arr[mid]==target)
    {
    return 1;
    }
    else if(arr[mid]>target)
    {
    return BS(arr,l,mid-1,target);
    }
    else
    {
    return BS(arr,mid+1,r,target);
    }
}
```

OUTPUT:



